

SMC5/6-Mediated Plasmid Silencing is Directed by SIMC1-SLF2 and Antagonized by the SV40 Large T Antigen

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eLife Assessment

This Research Advance manuscript further elucidates the roles of SMC5/6 loader proteins and associated factors in the silencing of extrachromosomal circular DNA by the SMC5/6 complex. While the findings are largely in line with expectations, they are **valuable**, representing a meaningful advance beyond the recent study from the same laboratories (PMC9708086), validating the previous model that distinct SMC5/6 subcomplexes, SIMC1-SLF2 and SLF1/2, separately control its transcriptional repression and DNA repair activities on extrachromosomal DNA. **Solid** evidence is presented for a role for SIMC1/SLF2 in localization of the SMC5/6 complex to plasmid DNA, and the distinct requirements as compared to recruitment of SMC5/6 to chromosomal DNA lesions.

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Abstract

SMC5/6 is unique amongst the Structural Maintenance of Chromosomes (SMC) complexes in its ability to repress transcription from extrachromosomal circular DNA (ecDNA), including viral genomes and plasmids. Previously, we showed that human SMC5/6 is regulated by two mutually exclusive subcomplexes—SIMC1-SLF2 and SLF1/2—the counterparts of yeast Nse5/6 (Oravcová, eLife, 2022). Notably, only SIMC1-SLF2 recruits SMC5/6 to SV40 Large T antigen (LT) foci in PML nuclear bodies (PML NBs), suggesting that these regulatory subcomplexes direct distinct roles of SMC5/6 on chromosomal versus ecDNA. However, their roles in plasmid repression remain unclear. Here, we demonstrate that SMC5/6-mediated repression of plasmid transcription depends exclusively on SIMC1-SLF2, whereas SLF1/2 is dispensable. Reinforcing its specialized role in ecDNA suppression, SIMC1-SLF2 does not participate in SMC5/6 recruitment to chromosomal DNA lesions. We further show that plasmid silencing requires a conserved interaction between SIMC1-SLF2 and SMC6, mirroring the functional relationship observed between yeast Nse5/6 and Smc6. As for viral silencing, plasmid repression depends on the SUMO pathway; however, unlike viral silencing, it does not require PML NBs. Additionally, we find that LT interacts with SMC5/6 and increases plasmid

transcription to levels observed in SIMC1-SLF2-deficient cells—echoing the antagonistic roles of HBx (HBV) and Vpr (HIV-1) in viral genome repression. These findings expand the paradigm of viral antagonism against SMC5/6-mediated silencing, positioning LT as a novel player in this evolutionary tug-of-war.

Introduction

The Structural Maintenance of Chromosomes (SMC) complexes, present in both prokaryotes and eukaryotes, are core modulators of chromosome architecture and organization, supporting the development and homeostasis of all living organisms (1,2). The SMC1/3 (cohesin) and SMC2/4 (condensin) complexes ensure accurate genome propagation by folding chromosomes into distinct structures. Meanwhile, SMC5/6 plays a key role in maintaining genome stability and integrity, contributing to processes such as homologous recombination (HR), alternative lengthening of telomeres (ALT), and ribosomal DNA (rDNA) replication (3–7). Recently, SMC5/6 has also been recognized as a viral restriction factor, highlighting its broader role in managing extrachromosomal DNAs (ecDNAs, (8–10)).

Beyond viral restriction, the repression of transcription from simple plasmid DNA has also been added to the activities of the human SMC5/6 “Swiss Army knife” of DNA manipulation (8–12). SMC5/6 selectively represses the transcription of circular but not linear plasmid DNA (12), which is likely related to their topological states. That is, DNA that accumulates superhelical stress during transcription is the preferred binding substrate for SMC5/6, whereas linear DNA that dissipates such tension is not stably bound by the complex (12–15). Once DNA bound, SMC5/6 may further compact and thereby restrict transcriptional activity on the ecDNA.

All SMC complexes share a fundamental structure consisting of a SMC heterodimer connected by a kleisin protein. These complexes form a ring-like structure that traps DNA. The SMC proteins have a bent-arm shape and can open to encircle DNA when their ATPase head domains bind ATP, aiding DNA movement through the ring. Kleisin plays a critical role for connecting and locking the SMC heads together (6,16–22). The SMC5/6 complex is particularly adept at recognizing and compacting special DNA structures, such as HR intermediates and supercoiled DNA (13,15,23,24).

The SMC5/6 complex is composed of a highly conserved “core” hexamer: the SMC5/6 heterodimer and non-SMC-elements (NSE) 1–4 (9). A Kleisin (Nse4 in yeast; NSMCE4 in human) and two interacting partners (Nse1 and Nse3 in yeast; NSMCE1 and NSMCE3 in human) form a trimer that joins the SMC5/6 head domains. The SUMO ligase Nse2 (NSMCE2 in human) attaches to the SMC5 arm. The complex’s functional specificity is further defined by non-core subunits, the Nse5/6 heterodimer (9,18,25).

Yeast Nse5/6 plays key roles in regulating SMC5/6’s genome stability functions, facilitating its chromatin loading, inhibiting its ATPase activity, and enhancing Nse2 SUMO ligase activity (19,26–32). Despite its critical regulatory roles, the identification of a human Nse5/6-like complex lagged, largely due to the lack of cross-species protein sequence conservation. Nevertheless, identified in proteomic approaches, the human SLF1 (SMC5-SMC6 Complex Localization Factor 1)-SLF2 heterodimer (SLF1/2) was shown to fulfill some genome stability roles performed by Nse5/6 in yeast (7). Interestingly, however, with the possible exception of HIV-1 proviral DNA, the repression of viral DNA transcription by SMC5/6 depends on SLF2 but not SLF1 (8,9,11). Therefore, SLF2 either functions as a monomer in viral repression, or there is a cofactor that replaces SLF1 in this role. Towards resolving this issue, we recently discovered SIMC1 as a missing link in SMC5/6 regulation and demonstrated that the SIMC1-SLF2 complex is a human counterpart of yeast Nse5/6 (33).

Our structural and mutational analyses revealed that SIMC1 and SLF2 contain Nse5 and Nse6-like domains, respectively, which interact through a conserved interface. SIMC1-SLF2 interacts with and directs SMC5/6 to subnuclear compartments at PML nuclear bodies (PML NBs) induced by the SV40 polyomavirus large T antigen (LT). Further, we showed that SIMC1-SLF2 and SLF1/2 are mutually exclusive subcomplexes of SMC5/6 that likely support the distinct antiviral and DNA repair activities of the holocomplex. Thus, SIMC1-SLF2 and SLF1/2 act as Nse5/6-like regulatory elements that modulate the broad activities of the SMC5/6 complex (33 [↗](#)).

Despite the foregoing advances, the specific roles of the SLF2-based SIMC1 and SLF1 subcomplexes in SMC5/6 regulation remained unclear. Although we previously proposed distinct functions for SIMC1-SLF2 and SLF1/2 in ecDNA silencing and DNA repair, respectively, these mechanisms had yet to be demonstrated. Additionally, SUMOylation, PML NBs, and SV40 LT were implicated in the regulation of SMC5/6-based ecDNA silencing, but their actual contributions remained untested. Here, we address these gaps directly. In line with our prior model, we show that SIMC1-SLF2, but not SLF1/2, mediates plasmid silencing through a direct and evolutionarily conserved interaction with SMC6. Conversely, SLF1/2 specifically recruits SMC5/6 to DNA lesions. Furthermore, we demonstrate that plasmid silencing is facilitated by the SUMO pathway but, unlike the SMC5/6-mediated silencing of HBV (34 [↗](#)), does not require PML NBs. Lastly, we reveal that SV40 LT binds SMC5/6 and antagonizes plasmid silencing, thereby adding polyomaviruses to the growing list of pathogenic viruses that counteract SMC5/6-based transcriptional repression.

Results

SIMC1-SLF2 associates with SMC6 through a conserved patch

Our previous structural analysis revealed that the Nse5-like domain of SIMC1 (SIMC1^{Nse5}) and the Nse6-like domain of SLF2 (SLF2^{Nse6}) both adopt α -solenoid-like structures (33 [↗](#)). These domains interact in a head-to-tail orientation, forming an ellipsoid-shaped dimer (Figure 1A [↗](#)). This dimerization clusters conserved residues from the N-terminus of SIMC1^{Nse5} and the C-terminus of SLF2^{Nse6} at one edge of the ellipsoid (Figure 1B [↗](#)). We previously used a SIMC1 mutant harboring four substitutions in α 1 of SIMC1 (SIMC1^{combo}, Figure 1B [↗](#)) to demonstrate that this conserved patch of SIMC1 is essential for binding to SMC6 and for the localization of the SMC5/6 complex at LT-containing PML NBs (33 [↗](#)).

In this study, we extended our analysis to SLF2 by introducing two sets of mutations into the Nse6-like region of SLF2 (SLF2^{CTD}; residues 635-1173): **mut 1** (E1121A/K1122A/K1125D/C1126R/E1130K in α 16) and **mut 2** (T1138R/K1141E/D1142R/A1145R/G1149E in the subsequent loop and α 17). These mutations were designed to disrupt charge interactions, hydrogen bonds, and van der Waals contacts, following the same strategy used with SIMC1^{combo} to disrupt binding (33 [↗](#)). Using HEK293 cells co-expressing FLAG-tagged SLF2^{CTD} mutants, GFP-tagged SIMC1 and Myc-tagged SMC6, we performed GFP pull-downs followed by Western blot analysis (Figure 1C [↗](#)). Wild-type SLF2^{CTD} interacted with both SIMC1 and SMC6 (Lane 1), whereas both **mut 1** and **mut 2** disrupted SMC6 binding while preserving the interaction with SIMC1 (Lanes 2, 3).

We also examined several control mutants. To confirm the SIMC1-SMC6 interface, we introduced a new SIMC1 control mutant containing four substitutions (Q842A/H846A/K849E/D857R) in α 17, located at the opposite end of the ellipsoid, away from the predicted SMC6 contact surface (Figure 1A [↗](#)). This SIMC1 mutant bound to SLF2 but more weakly than wild-type SIMC1 (Figure 1—figure supplement 1A [↗](#)). This was an unexpected finding that is difficult to rationalize based on the structure. However, despite the reduced SLF2 association, it fully retained its interaction with SMC6, indicating that the residues mutated in SIMC1^{combo} are specific to the SMC6 interface. In this experiment using overexpressed proteins, SIMC1 and SMC6 appear capable of maintaining their association during the pull-down even without stoichiometric SLF2. We then evaluated the

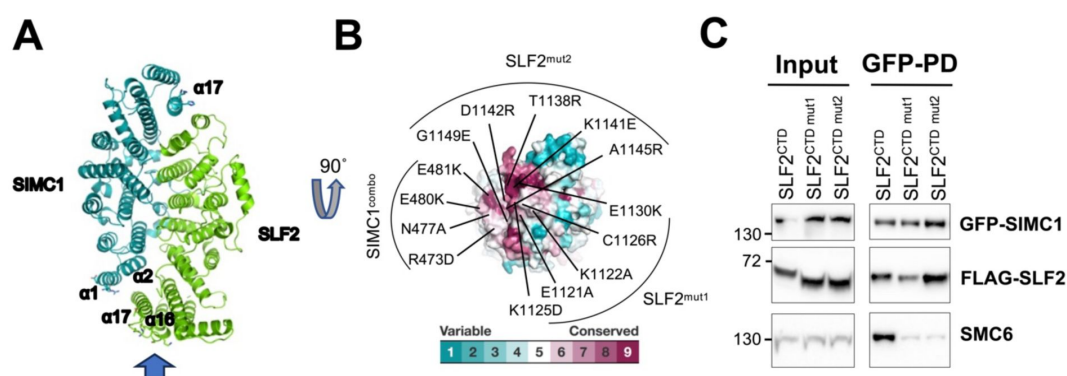


Figure 1

SIMC1-SLF2 composite surface patch is required for SMC6 association.

(A) The cryo-EM structure of the SIMC1-SLF2 complex (PDB ID: 7T5P) ([33](#)). Residues mutated for interaction analysis are shown with side chains. The arrow at the bottom indicates the viewing angle for (B). **(B)** Conservation mapping on the surface of SIMC1-SLF2 complex, showing the N-terminus of SIMC1 and the C-terminus of SLF2. Conservation scores obtained from the Consurf server ([35](#)) are shown by the color graduation as indicated. Amino acids mutated in SIMC1^{combo} (R473D/N477A/E480K/E481K in $\alpha 1$ of SIMC1), SLF2 mut1 (E1121A/K1122A/K1125D/C1126R/E1130K in $\alpha 16$ of SLF2) and SLF2 mut2 (T1138R/K1141E/D1142R/A1145R/G1149E in $\alpha 17$ of SLF2) are labeled. **(C)** Western blot of GFP-Trap immunoprecipitation from HEK293 cells transfected with plasmids expressing GFP-SIMC1, Myc-SMC6, and FLAG-tagged C-terminal domain (CTD) of SLF2, either WT or containing mutations (mut1, mut2). Input and GFP PD were detected with anti-GFP, FLAG, or SMC6 antibody.

most C-terminal residues of SLF2 (1161–1173), which are unstructured. Despite their conservation, deleting these residues (SLF2⁶³⁵⁻¹¹⁶⁰) did not affect SMC6 binding (**Figure 1—figure supplement 1B**). Collectively, these results confirm that the conserved surface patch of SIMC1-SLF2 is essential for SMC6 binding.

SIMC1-SLF2 subcomplex contacts the neck region of SMC6

We used AlphaFold-Multimer (36) to predict the structure of the SIMC1-SLF2-SMC6 complex (**Figure 2A**, **Figure 2—figure supplement 1A**). The resulting high-confidence model, as assessed by the predicted aligned error (PAE) plot (**Figure 2—figure supplement 1B**), aligns well with mutational data and highlights the conserved surface patch of SIMC1-SLF2 as the interface with SMC6 (**Figure 2A**). SMC6 is a long, hairpin-like polypeptide in which its N- and C-termini fold together to form the ATPase “head” domain at one end. The “hinge” domain at the bending end mediates heterodimerization with SMC5 (**Figure 2—figure supplement 1A**). Connecting the head and hinge is a flexible coiled-coil; the portion adjacent to the head is referred to as the “neck” (**Figure 2A**).

In the model, the conserved SIMC1-SLF2 surface patch interacts with C-terminal residues of SMC6, centered on the neck and opposite the ATPase active site where it dimerizes with SMC5’s ATPase domain (21,30). This interface shows excellent shape complementarity—SMC6’s convex neck helix fits snugly into the concave surface formed by $\alpha 1$ and $\alpha 2$ of SIMC1 and $\alpha 17$ of SLF2 (**Figure 2A**). Additional interactions arise from SLF2’s $\alpha 16$ and the $\alpha 16$ - $\alpha 17$ loop with SMC6’s head residues, spanning a total buried surface area of 1516 Å².

This model explains the roles of the amino acids altered in our binding studies (**Figure 2B**; **Figure 2—figure supplement 1C, D**). The four residues modified in SIMC1^{combo} all make direct contact with SMC6’s neck helix, including three salt bridges: R473^{SIMC1}-D946^{SMC6}, E480^{SIMC1}-K957^{SMC6}/K942^{SMC6}, and E481^{SIMC1}-K942^{SMC6}. The SLF2 mutations also engage SMC6’s neck region, forming salt bridges (E1130^{SLF2}-R940^{SMC6} and K1141^{SLF2}-E1008^{SMC6}), a cation- π interaction (K1125^{SLF2}-Y944^{SMC6}), and multiple van der Waals contacts (**Figure 2A**; **Figure 2—figure supplement 1C, D**). Thus, the model and mutational data are consistent.

Next, we validated these SMC6’s interface contacts using alanine substitutions. We introduced three sets of mutations: 1) **SIMC1-facing** residues (K942A/D946A/K957A), which form salt bridges with SIMC1; 2) **SLF2-facing** residues (R940A/Y944A/N947A), which form salt bridges and polar and hydrophobic contacts with SLF2; 3) **SIMC1-SLF2-groove-facing** (L939A/R940A/K942A/L943A), which alters two leucines that insert into a hydrophobic groove between $\alpha 1$ of SIMC1 and $\alpha 17$ of SLF2, together with the SIMC1- and SLF2-facing residues (K942 and R940, respectively) (**Figure 2B**). When these SMC6 mutants were co-expressed with SIMC1 and SLF2, pull-down assays showed significantly reduced or abolished binding (**Figure 2C**, lane 2-4). Thus, our mutational data strongly support the AlphaFold model.

We conclude that SIMC1-SLF2 associates with the neck region of SMC6, opposite the ATPase active site used for heterodimerization with SMC5 (**Figure 2D**; **Figure 2—figure supplement 1A**). A recent cryo-EM study on the *S. cerevisiae* Smc5/6 complex similarly demonstrated that Nse5/6 engages with the neck region of Smc6 (30), reinforcing the idea that this mode of interaction is evolutionarily conserved (**Figure 2D**). This has implications for the function of SIMC1-SLF2 in the regulation of SMC5/6 activity, based on the findings for yeast Nse5/6 (see discussion).

SMC5/6 mediated plasmid silencing requires the Nse5/6-like SIMC1-SLF2 complex

To examine the role of SIMC1 and SLF2 in SMC5/6-mediated episomal DNA silencing, we used CRISPR-Cas9 to create U2OS SIMC1^{-/-} or SLF2^{-/-} cells. We first compared the expression of a transfected GFP reporter plasmid between wild-type (WT) and null cells. GFP transcripts, as

quantified by RT-qPCR, showed a strong increase over WT in both SIMC1^{-/-} and SLF2^{-/-} cells (**Figure 3A**). This increase in GFP transcripts was not due to an increase in average plasmid copy number in SLF2 null cells, versus WT, as determined by qPCR (**Figure 3—figure supplement 1A**). We then used fluorescence activated cell sorting (FACS) to monitor GFP expression (37) and found that both SIMC1^{-/-} and SLF2^{-/-} cells expressed significantly more GFP than WT (**Figure 3B**). Importantly, expressing WT SIMC1 or SLF2 in the corresponding null cells reduced reporter expression to levels observed in WT cells (**Figure 3B**), demonstrating the involvement of SIMC1-SLF2 in plasmid silencing. Notably, however, SLF2^{mut1} that abolishes the SIMC1-SLF2 interaction with SMC6 (**Figure 1C**) did not reduce reporter expression in SLF2^{-/-} cells, despite being well expressed (**Figure 3B—figure supplement 1B**). Importantly, in keeping with related studies (38–40), we found that the effect of SLF2 on GFP expression was lost upon integration of the GFP reporter plasmid into the genome (**Figure 3—figure supplement 1C**). Together, these findings demonstrate that SIMC1-SLF2, and their interaction with SMC6, are required for the selective repression of episomal plasmid DNA.

SIMC1 and SLF1 direct SMC5/6 to plasmid silencing and DNA repair, respectively

We previously showed that SLF1/2 and SIMC1-SLF2 form mutually exclusive complexes with SMC5/6 and recruit it to distinct cellular locations (33). Specifically, epitope-tagged SLF1 colocalized with gamma-H2AX at laser stripes, whereas SIMC1 did not. Here we more directly examined the roles of SLF1 and SIMC1 in recruiting SMC5/6 to DNA damage sites. To this end, we induced localized DNA lesions using laser microirradiation in WT, SLF1^{-/-}, and SIMC1^{-/-} U2OS cells. In WT and SIMC1^{-/-} cells SMC5/6 accumulated at laser-induced lesions marked by gamma-H2AX, whereas no such accumulation was seen in SLF1^{-/-} cells (**Figure 4A**). Thus, SLF1 but not SIMC1 plays a critical role in directing SMC5/6 to DNA damage sites. Conversely, when comparing GFP reporter expression in WT and SLF1 null cells, we found no change in GFP levels in the absence of SLF1 (**Figure 4B**). Thus, unlike SIMC1, SLF1 does not play a role in SMC5/6-mediated plasmid silencing, which is consistent with what has been observed in SLF1^{-/-} HepG2 cells (10).

SUMOylation supports plasmid repression

Given that SIMC1 contains SUMO interacting motifs (SIMs), and the SMC5/6 complex is a SUMO ligase that contains other SUMO-interacting interfaces, we also tested for a potential involvement of the SUMO pathway in plasmid silencing (33,41,42). Cells were treated with the SUMO pathway inhibitor (SUMOi), TAK-981 (43). We observed a strong dose-dependent increase in reporter gene expression in both U2OS and RPE cells (**Figure 5—figure supplement 1A, B**), supporting an involvement of the SUMO pathway in plasmid silencing. This is consistent with the broadly observed transcriptionally repressive activity of SUMOylation (44,45). Interestingly, SUMOi treatment does not significantly increase the level of GFP reporter expression in SIMC1^{-/-} and SLF2^{-/-} over its effect on WT, despite the pre-existing defect in plasmid silencing in SIMC1 and SLF2 null cell lines (**Figure 5A**). The lack of additivity between SUMOi and SIMC1^{-/-} or SLF2^{-/-} suggests that SMC5/6-mediated plasmid silencing operates in large part under the broad “umbrella” of SUMOylation-based transcriptional repression. Intriguingly, when the GFP reporter plasmid was integrated into the genome, its expression was unaffected by SUMOi treatment in either WT or SLF2^{-/-} U2OS cells (**Figure 5—figure supplement 1C**). This was somewhat unexpected given the broadly inhibitory effect of SUMO on transcription. One possibility is that the dynamic range of transcriptional regulation is larger for episomal DNA, resulting in the muted response of the integrated reporter. This phenomenon merits further investigation.

Figure 3

SIMC1/SLF2-SMC6 interaction is critical for plasmid silencing.

(A) Expression of a GFP reporter transiently transfected into U2OS WT and SIMC1^{-/-} or SLF2^{-/-} cell lines was measured by reverse transcription-quantitative PCR 72 hours after transfection. GFP expression was normalized to the expression of beta-actin (mean ± s.d. from n=4 independent experiments, two-tailed unpaired t-test; (**) p<0.005). (B) U2OS WT, SIMC1^{-/-} or SLF2^{-/-} cells with integrated empty vector or vector expressing SIMC1 or SLF2 variants (WT, mut1), respectively, were transiently transfected with GFP reporter. After 72 hours, GFP intensity was measured by FACS and is displayed relative to GFP intensity measured in WT cells. Data are the average ± s.d. from n=3 independent experiments, two-tailed unpaired t-test; (**) p<0.005. No significant difference (p>0.05) was found between WT and SIMC1^{-/-} + pSIMC1 (left panel), or between WT and SLF2^{-/-} + pSLF2 (right panel).

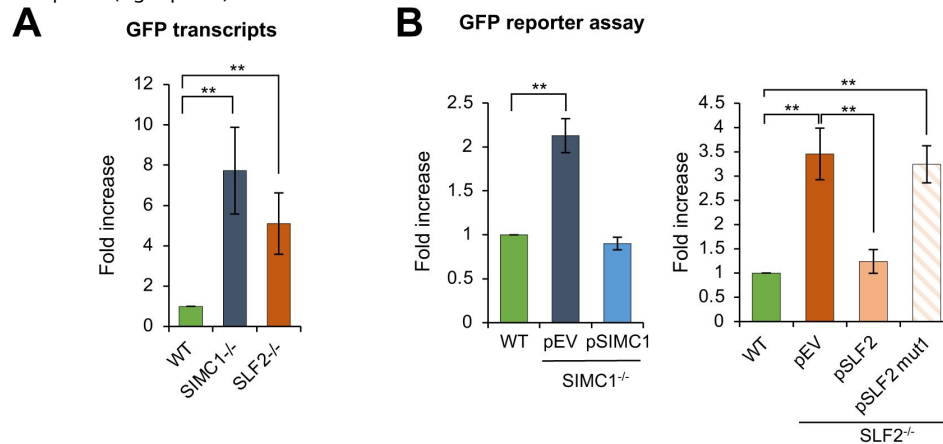
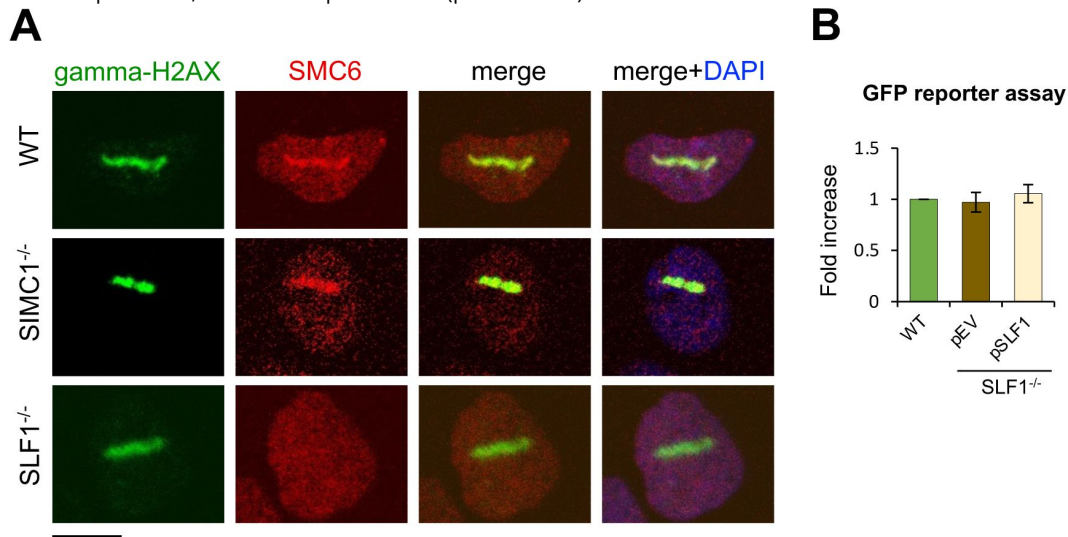


Figure 4

SLF1 is involved in SMC5/6 DNA damage repair, not plasmid silencing.

(A) Representative immunofluorescence images of U2OS WT, SIMC1^{-/-}, SLF1^{-/-} cells exposed to laser microirradiation upon treatment with 50 uM angelicin and 1 ug/ml Hoechst 33342 for 30 min. One hour after microirradiation, cells were pre-extracted, fixed and stained with gamma-H2A.X (phosphorylation of histone H2A.X Ser139; green), SMC6 (red) antibodies and DAPI (blue). Scale bar 10 µm. (B) GFP intensity was measured by flow cytometry 72 hr after reporter transient transfection in U2OS WT cells or SLF1^{-/-} cells with integrated empty vector or stably expressing SLF1. Data are the average ± s.d. from n=3 independent experiments, two-tailed unpaired t-test (p-value >0.05).



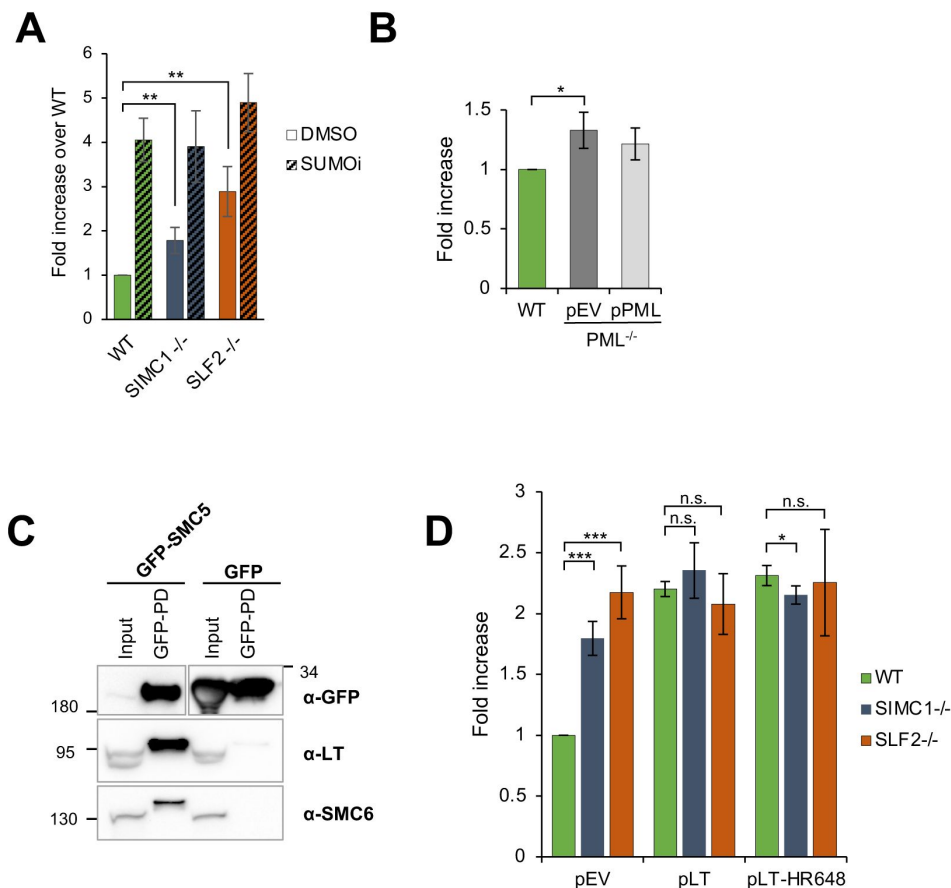


Figure 5

Plasmid silencing depends on the SUMO pathway and LT, not PML NBs.

(A) U2OS WT, SIMC1^{-/-} or SLF2^{-/-} cells were transiently transfected with GFP reporter and treated with 100 nM TAK-981(SUMOi) or DMSO at the same time. After 72 hours, GFP intensity was determined by FACS. Data are normalized to WT cells treated with DMSO and represent the average $\pm$ s.d. from n=4 independent experiments, two-tailed unpaired t-test; (**) p<0.005. No significant difference was found between WT and SIMC1^{-/-} or SLF2^{-/-} when treated with SUMOi (p>0.05). **(B)** U2OS WT or PML^{-/-} cells with integrated empty vector or vector expressing PML, respectively, were transiently transfected with GFP reporter. Intensity of GFP was measured by FACS 72 hours after transfection. Data are the average $\pm$ s.d. from n=3 independent experiments, two-tailed unpaired t-test; (*) p<0.05. No significant difference (p>0.05) was found between WT and PML^{-/-} + pPML. **(C)** Western blot of GFP-trap immunoprecipitation from HEK293 cells transiently transfected with either GFP-SMC5 or GFP alone in combination with SV40 vector expressing LT and Myc-SMC6. Signals were visualized using GFP, LT and SMC6 antibodies. **(D)** GFP intensity measured by FACS in U2OS WT, SIMC1^{-/-} or SLF2^{-/-} cells with integrated empty vector, vector expressing large T antigen (LT) or LT-HR684 variant, respectively that were transiently transfected with GFP reporter for 72 hours. Data are the average $\pm$ s.d. from n=4 independent experiments. (*) p<0.05; (***) p<0.0005; (n.s.) p>0.05 (two-tailed unpaired t-test).

PML NBs are not required for plasmid silencing

We previously showed that SIMC1-SLF2 directs SMC5/6 to LT antigen-containing SV40 replication centers at PML NBs (33). Moreover, SMC5/6-mediated restriction of HBV occurs at PML NBs (34,46) and, the localization of SMC5/6 at PML NBs is SLF2-dependent in HepG2 and PHH cells (10,46). We also now find that SLF2 is required for the localization of SMC5/6 to PML NBs in U2OS cells (Figure 5—figure supplement 2). Based on the foregoing, PML NBs have been proposed, but not directly shown, to support simple plasmid repression (10). Therefore, we tested the impact of PML NBs on plasmid silencing by measuring GFP reporter expression in WT and PML^{-/-} U2OS cells (47). Compared to SIMC1 and SLF2 null cells, we observed only a minor increase in GFP expression in PML^{-/-} versus WT cells (Figure 5B). Moreover, the stable re-expression of PML in PML^{-/-} cells did not significantly reduce the level of GFP expression, as compared to vector control, suggesting the slight increase observed is due to clonal variation from CRISPR editing (Figure 5B). Thus, we conclude that PML NBs are not needed for the SMC5/6-mediated silencing of plasmids.

SV40 Large T antigen (LT) overcomes plasmid silencing

Considering that HBx and Vpr antagonize SMC5/6 to induce HBV and HIV-1 transcription, respectively (38,39,48), we tested if SV40 LT similarly impacts plasmid silencing. In this regard, we previously found that SMC5/6 colocalizes with LT-induced nuclear foci at PML NBs in a SIMC1-SLF2-dependent manner (33). Here, we show that ectopically expressed LT coimmunoprecipitates with GFP-SMC5/6 but not a GFP control (Figure 5C). Therefore, LT, like HBx and Vpr, may contact SMC5/6 to antagonize its ecDNA repression activity.

Indeed, the stable expression of LT in WT cells promoted GFP reporter plasmid expression, as compared to WT cells with an empty vector control (Figure 5D; Figure 5—figure supplement 3A). The host range defective mutant LT-HR684 had a similar effect as WT LT, indicating that the C-terminus of LT, which inhibits the SV40 host range restriction factor FAM111A (49), is not involved in overcoming plasmid repression (Figure 5D). Notably, the enhancement of plasmid DNA transcription by LT is not additive with that already present in SIMC1^{-/-} and SLF2^{-/-} cells (Figure 5D). The epistatic relationship between the positive effects of SIMC1^{-/-}, SLF2^{-/-}, and LT expression on plasmid transcription indicates that LT overcomes SMC5/6-mediated repression. Interestingly, as we observed with SUMOi, when the GFP reporter plasmid was integrated in WT or SLF2^{-/-} U2OS cells, there was not a pronounced effect of LT on GFP expression (Figure 5—figure supplement 3B). This is like the selective effect of HBx, whereby it overcomes the innate repression of transcription from episomal but not chromosomal reporters (40).

Discussion

Our structural analyses indicate that SIMC1-SLF2 engages the SMC5/6 core at the neck region of SMC6, situated opposite the ATPase active site. These findings, together with recent cryo-EM analyses of the *S. cerevisiae* Smc5/6 complex (20,21,31), highlight a conserved mode of interaction in which Nse5/6 or Nse5/6-like subunits bind to the backside of SMC6's head domain. This binding site is ideal for allosteric regulation of the ATPase and DNA-binding functions of the SMC5/6 family. Indeed, yeast Nse5/6 reduces Smc5/6 ATPase activity and disrupts the Nse4 (kleisin)-Smc6 interface, which may open the gate to enable topological DNA binding by Smc5/6 (Figure 6, (20,30,31)). The clear overlap between our validated SIMC1-SLF2-SMC6 structure and that of yeast Nse5/6-Smc6 suggests that SIMC1-SLF2, and its orthologue SLF1/2, may similarly modulate SMC5/6 ATPase and DNA binding activities, albeit at distinct DNA targets.

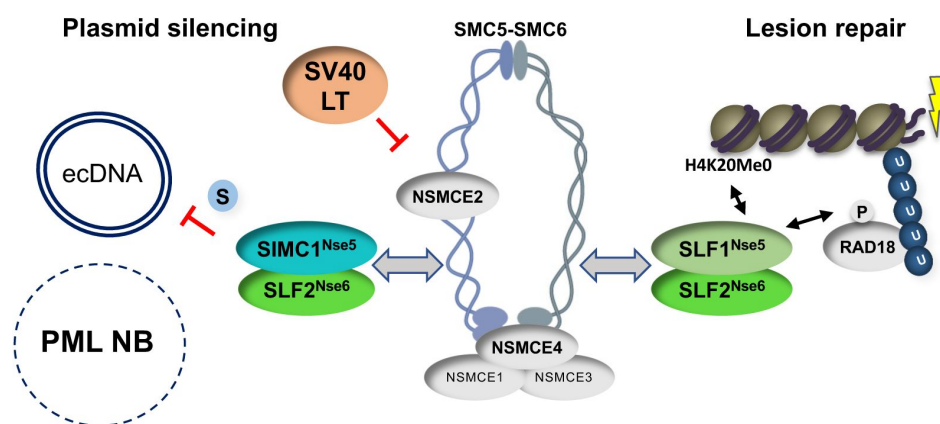


Figure 6

SIMC1-SLF2 and SLF1/2 subcomplexes direct SMC5/6 to separate pathways of plasmid silencing and DNA lesion repair.

SIMC1-SLF2 and SMC5/6-mediated plasmid silencing is facilitated by the SUMO pathway and antagonized by SV40 LT. PML NBs are not involved in plasmid silencing. S stands for SUMO; U for ubiquitin. Both SIMC1-SLF2 and SLF1/2 likely contact SMC6 directly through their conserved Nse5/6-like domains, but the SIMC1 SIMs and SLF1 BRCT/ARD domains contact distinct posttranslational modifiers to direct the complex to its separate roles.

The distinct targeting functions of the SIMC1 and SLF1-based SLF2 complexes is underscored by our current analysis of microirradiated cells that lack each complex (**Figure 4**). While the deletion of SLF1 abrogates SMC5/6 localization at DNA lesions, the loss of SIMC1 has no effect (**Figure 4A**). On the other hand, SIMC1 is required for the recruitment of SMC5/6 to LT-containing PML NBs (33). Lastly, we found that SLF1 does not impact plasmid silencing (**Figure 4B**), consistent with several analyses of HIV-1, rAAV, and HBV silencing (10,39,46,50) but contrasting with another study (51). It remains unclear why this latter discrepancy exists. While there may still be overlapping functions of SIMC1 and SLF1 to uncover at the organismal level, it is clear they support distinct cellular responses to ecDNAs and DNA lesions (**Figure 6**).

In keeping with the SUMO-binding ability of SIMC1 (**Figure 6**, (42)), we found that the inhibition of SUMOylation by TAK-981 strongly enhanced plasmid transcription (**Figure 5-figure supplement 2**). This is likely a broad effect that includes the loss of repressive modifications of transcription factors (45); as well as the lifting of SMC5/6-mediated repression to enhance ecDNA transcription. A similar effect of TAK-981 was seen on HIV-1 proviral DNA gene expression, which was elevated by SUMO inhibition to a similar degree as that observed in SMC5Δ cells (51). The authors concluded that the effect of TAK-981 was entirely through the inhibition of SMC5/6-mediated SUMOylation (NSE2 subunit). This is different to our results, which show that SUMOylation inhibition has a larger effect than the loss of the SIMC1-SLF2 complex on plasmid gene expression. The distinct natures of the ecDNAs under study, plasmid versus HIV-1 proviral DNA, and the genetic backgrounds may contribute to this difference. Nevertheless, our data support a key role of the SUMO pathway in plasmid repression, and that SMC5/6 works within it.

Despite the fact that SIMC1-SLF2 localizes SMC5/6 to PML NBs that contain SV40 LT (31), and HBV is restricted at PML NBs (34), they are dispensable for plasmid silencing. While the reason for the different impact of PML NBs in HBV versus simple plasmid silencing remain unknown, it may be related to the natural localization of each ecDNA. That is, HBV localizes to PML NBs as part of the viral life cycle, in a manner dependent on the SUMOylation of its core protein (52). In contrast, the direct visualization of plasmids revealed they are pan-nuclear and apparently excluded from PML NBs (53). It is also noteworthy that a recent genome-scale CRISPR screen for repressors of rAAV transcription identified all components of the SMC5/6 complex, including SIMC1 and SLF2, but not SLF1 or PML (50). This raises interesting questions about the mechanism of SMC5/6-mediated transcriptional silencing, since it can occur outside PML NBs. In this regard, SUMOylation was recently shown to support the recombination-based alternative lengthening of telomeres (ALT) pathway in the absence of PML NBs (54), which are normally key components of the fully active pathway (47). Therefore, the SUMOylation of plasmid associated proteins, and or SMC5/6, may support transcriptional repression of plasmids in the absence of PML NBs, consistent with our findings using TAK-981.

Interestingly, the expression of LT enhances plasmid transcription in a non-additive manner with the deletion of SIMC1 or SLF2, suggesting an epistatic relationship. While LT is known to be a promiscuous transcriptional activator (55,56) that does not rule out a co-existing role in antagonizing SMC5/6. Indeed, these findings are reminiscent of HBx from HBV and Vpr of HIV-1, both of which are known promiscuous transcriptional activators that also directly antagonize SMC5/6 to relieve transcriptional repression (12,38,39,48,57,58–59). Unlike HBx and Vpr, the interaction of LT with SMC5/6 does not appear to induce the degradation of its subunits (33). Nevertheless, LT binds and sterically inhibits several cellular proteins instead of inducing their degradation, including Rb and p53 (60). Further analysis will be needed to define LT-SMC5/6 contacts, and its potential mode of inhibition. This will have important ramifications for the lifecycle of the pathogenic polyomaviruses since their LT proteins share functions with SV40 LT (61).

In conclusion, our structure-function analyses delineate a specific function for the SIMC1-SLF2 complex in SMC5/6-mediated plasmid repression. This resolves a discrepancy in the field, wherein SLF2 was thought to act alone in ecDNA repression, since SLF1 is not required. While further studies are needed, SIMC1-SLF2 is most likely responsible for supporting SMC5/6-mediated ecDNA silencing, including that of multiple pathogenic viruses. Although HBV silencing requires PML NBs, we find that plasmid silencing does not. This represents a departure from current dogma, suggesting that the SMC5/6-mediated repression mechanism does not always rely on PML NB-resident factors. In some cases, the SUMOylation of ecDNA-associated proteins may be sufficient to support SMC5/6-based repression. Finally, because the SV40 LT antigen interacts with SMC5/6 and antagonizes its repression of plasmids, the orthologous polyomaviral LT antigens are likely to block SMC5/6-mediated restriction of pathogenic polyomaviruses.

Materials and Methods

Construction of recombinant plasmids

Construction of plasmid DNA was described in (33 [DOI](#)). Mutations in SIMC1 (combo ctrl), SLF2 (mut1, mut2) and SMC6 (SIMC1-facing, SLF2-facing, SIMC1-SLF2-groove-facing) were either introduced in primers or a gBlock sequence containing desired mutations was purchased (IDT). All plasmids created in this study have been verified by sequencing service provided by Plasmidsaurus. Additional details of plasmid construction are available upon request.

Cell culture, transfection, stable line generation

All cell lines and their derivatives were cultured in DMEM (Gibco, #11995065) supplemented with 10% (v/v) fetal bovine serum (Omega Scientific, #FB-01), 1% (v/v) antibiotic-antimycotic (Gibco, #15240062) and maintained at 37 °C in humidified air with 5% CO₂.

Transient plasmid transfections and generation of stable cell lines was detailed in (33 [DOI](#)). A list of cell lines and vectors used for generating stable cell lines or transient transfection are listed below:

cell line (<i>Homo-sapiens</i>)			
Designation	Source or reference	Identifier	Additional information
HEK293	DOI: 10.7554/eLife.79676		
HEK293T; U2OS	DOI: 10.7554/eLife.79676		

U2OS SIMC1 ^{-/-}	this paper	clone B9	Derived from U2OS by CRISPR/ Cas9
U2OS SLF1 ^{-/-}	this paper	clone E9	Derived from U2OS by CRISPR/ Cas9
U2OS SLF2 ^{-/-}	this paper	clone 18	Derived from U2OS by CRISPR/ Cas9
U2OS PML ^{-/-}	DOI: 10.1101/gad.333963.119	clone 2C	
RPE	DOI: 10.1101/gad.333963.119		
Lentiviral vectors to generate stable cell line			
pHAGE2-FLAG-SIMC1	DOI: 10.7554/eLife.79676	pNB185	
pHAGE2	DOI: 10.7554/eLife.79676	pNB248	
pHAGE2-FLAG-SLF2	this paper	pNB525	
pHAGE2-FLAG-SLF2 mut1	this paper	pNB527	SLF2 mutations E1121A/ K1122A/ K1125D/ C1126R/ E1130K
pHAGE2-Flag-SLF1	this paper	pNB555	
pHAGE2-LT	this paper	pNB648	
pHAGE2-GFP	this paper	pNB670	
pHAGE2-LT-HR684	this paper	pNB717	
pMD2G	Addgene	#12259	
psPAX2	Addgene	#12260	
Vectors used for transient transfection			
pDEST-eGFP-NLS-STOP	Helle Ulrich lab	pNB68, pNZ110	

Dual secreted luciferase reporter	Addgene	#181934 pNB671	EF1alpha-Gaussia luciferase, CMV-Cypridina luciferase
pBlueScript KS-SV40	DOI: 10.1371/journal.ppat.1002949	pNB371	strain 776
pDEST-eGFP-SIMC1	DOI: 10.7554/eLife.79676	pNB263	
pDEST-FLAG-NLS-SLF2 (635-1173)	DOI: 10.7554/eLife.79676	pNB439	SLF2 ^{CTD}
pDEST-GFP-SIMC1 combo1 mut	DOI: 10.7554/eLife.79676	pNB497	SIMC1 mutations R473D/ N477A/ E480K/ E481K
pDEST-GFP-SIMC1 combo1 ctrl	this paper	pNB499	SIMC1 mutations Q842A/ H846A/ K849E/ D857R
pDEST-Myc-SMC6	DOI: 10.7554/eLife.79676	pNB530	
pDEST-FLAG-NLS-SLF2 (635-1160)	this paper	pNB535	
pDEST-FLAG-NLS-SLF2 (635-1173) mut1	this paper	pNB536	SLF2 mutations E1121A/ K1122A/ K1125D/ C1126R/ E1130K
pDEST-FLAG-NLS-SLF2 (635-1173) mut2	this paper	pNB537	SLF2 mutations T1138R/ K1141E/ D1142R/ A1145R/ G1149E
pDEST-GFP-SMC6	this paper	pNB546	
pDEST-GFP-SMC6 (groove-facing)	this paper	pNB677	SMC6 mutations L939A/ R940A/ K942A/ L943A
pDEST-GFP-SMC6 (SLF2-facing)	this paper	pNB678	SMC6 mutations R940A/ Y944A/ N947A
pDEST-GFP-SMC6 (SIMC1-facing)	this paper	pNB679	SMC6 mutations K942A/ D946A/ K957A
pDEST-2Myc-SIMC1	this paper	pNB680	
pDEST-Flag- LT wt	this paper	pNB739	

Cell line generation by CRISPR/Cas9

SIMC1, SLF1 and SLF2 knockout clones were generated via reverse transfection of CRISPR/Cas9 Ribonucleoprotein. A mixture of three sgRNAs specific for each gene (Synthego; 24 nM) was combined with 6 nM Cas9-NLS (Synthego) in reduced-serum medium OptiMEM (Life Technologies, #31985062) and incubated for 10 min followed by a 15 min incubation with TransIT-X2 (Mirus, #MIR6000). 6×10^4 cells were added and plated into 12-well plates. After 72 hours, cells were diluted and plated to isolate single cell clones that were genotyped to confirm successful editing. sgRNA sequences and diagnostic primers used in this study:

Target gene	Source	Identifier	Sequence 5'- 3'
gene knockout kit – SIMC1	Synthego	Guides for human SIMC1 CRISPR/Cas9	CCACAGGGACAAACTCTGCC; CTGCTGAAAGTCATCTTCTA; ATTGTGGGGCTGCTTGTAC
SIMC1 CRISPR diagnostic_F	this paper	oNB576	GGGCTTAGTATTTATGAGAGC
SIMC1 CRISPR diagnostic_R	this paper	oNB1023	TGAATCACTGCACCTGGTCT
gene knockout kit – SLF1	Synthego	Guides for human SLF1 CRISPR/Cas9	AGGAAAGTGGATACTAACCA; TTGATGAAACAACCTTATGAA; CGTGAAGAACTGAAACGCAC
SLF1 CRISPR diagnostic_F	this paper	oNB1024	TCCCCAAAATGCATAGTTCAAAG
SLF1 CRISPR diagnostic_R	this paper	oNB1025	ACCATGGCTCATTTGGGCTA
gene knockout kit – SLF2	Synthego	Guides for human SLF2 CRISPR/Cas9	ATGGTATACATGAGTCACGT; CTCCAAAAAGCAGACCACAG; AACTGGAATTTAGCTCCAG
SLF2 CRISPR diagnostic_F	this paper	oNB1190	CCCCAAAAGGGTGCCACCAGA
SLF2 CRISPR diagnostic_R	this paper	oNB1191	CTCCATTTGCTCCTTTCTCAACC

Co-immunoprecipitation and Western blotting

GFP-labeled proteins were immunoprecipitated using the Nano-Trap magnetic agarose (Chromotek) following the manufacturer's instructions. The protocol as well as whole cell lysate preparation for SDS-PAGE and western blot are described in detail in [\(33\)](#). Antibodies used for immunoblotting:

Antibody	Source	Identifier	Dilution
GFP-Trap magnetic agarose (Alpaca Monoclonal)	ChromoTek	gtma	IP: 25ul slurry
Myc-Trap magnetic agarose (Alpaca Monoclonal)	ChromoTek	ytma	IP: 25ul slurry
Anti-FLAG (Mouse monoclonal)	Sigma	F3165	1:5000
Anti-GFP (Mouse monoclonal)	Santa Cruz	sc-9996	1:10000
Anti-Myc (Mouse monoclonal)	Invitrogen	MA1-980	1:2000
Anti-PSTAIR (Mouse monoclonal)	Sigma	P7962	1:8000
Anti-SMC6 (Rabbit Polyclonal)	Bethyl	A300-237A	1:1000
Anti-SV40 LT (Mouse monoclonal)	Abcam	ab16879	1:5000
Goat Anti-Rabbit IgG, HRP (Goat Polyclonal)	Invitrogen	31460	1:5000
Goat Anti-mouse IgG, HRP (Goat Polyclonal)	Invitrogen	31430	1:5000

RNA extraction and quantitative PCR

For monitoring SLF2 and GFP expression by qPCR, cells were harvested for RNA isolation 72 hours after GFP vector transient transfection. Total RNA precipitated using RNeasy Plus Mini kit (Quiagen, #74104) was treated with DNaseI kit (Invitrogen, #18068-015) before cDNA synthesis by SuperScript III First-Strand Synthesis System for RT-PCR (Invitrogen, #18080-051). SensiFAST SYBR No-ROX kit (Meridian Bioscience, #BIO-98020) was used for qPCR and gene expression was calculated as $2^{-\Delta C_t}$ using beta-actin as a reference. Oligonucleotides used for RT-qPCR:

Target gene	Source	Identifier	Sequence 5'- 3'
SLF2_F	this paper	oNB390	GGAAGTCACTATTCATTC
SLF2_R	this paper	oNB476	GAATTTAGCTCCCAGTGGG
GFP_F	this paper	oNB1220	AGAAGAACGGCATCAAGG
GFP_R	this paper	oNB1221	GCTCAGGTAGTGGTTGTC
beta actin_F	DOI: 10.7554/eLife.79676	oNB492	AGGCACCAGGGCGTGAT
beta actin_R	DOI: 10.7554/eLife.79676	oNB493	GCCCACATAGGAATCCTTCTGAC

DNA extraction for qPCR

U2OS cells (1×10^5) with either integrated or transfected GFP vector (harvested 72 hours after GFP vector transfection) were resuspended in 100 μ l PBS and lysed in 200 μ l extraction buffer (0.1 M EDTA (pH 8.0), 0.5% (w/v) SDS, 10 mM Tris-Cl (pH 8.0)) supplemented with 40 μ g Proteinase K, 5 μ g RNaseA and incubated at 55 °C for 1 hour, vortexed occasionally. Lysate was sonicated on Bioruptor Pico (Diagenode) in 10 cycles of 15 s ON and 30 s OFF and cleared by centrifugation at 18,000 g 10 min. DNA was purified using NucleoSpin Gel and PCR clean-up kit (Macherey-Nagel, #740609) with 5 volumes of NTB binding buffer based on manufacturer's instructions. Oligonucleotides used for quantitative PCR:

Target gene	Source	Identifier	Sequence 5'- 3'
GFP_F	this paper	oNB1220	AGAAGAACGGCATCAAGG
GFP_R	this paper	oNB1221	GCTCAGGTAGTGGTTGTC
beta actin_F	this paper	oNB1378	AGAGCAAGAGAGGCATCC
beta actin_R	this paper	oNB1379	TTTCTCCATGTCGTCCCA

Flow cytometry

GFP fluorescence measurements were based on (37). GFP reporter vector was transiently transfected into cells using TransIT-LT1 (Mirus, #MIR2300). After 72 hours, cells were washed in PBS, dissociated using 0.25% trypsin and resuspended in DMEM. For SUMOi experiments, cells were treated with 100 nM TAK-981 (Cayman Chemical, #32741) or 0.1% DMSO (Sigma, #D8418) in control plates at the time of GFP vector transfection. Median GFP intensity was measured on Cytoflex S (Beckman Coulter) using the CytExpert software or on Aurora (Cytek) using SpectroFlo software. A minimum of 1100 GFP-positive cells were analyzed.

DNA damage by laser microirradiation

Protocol adapted from (62). U2OS cells (1.5×10^5) were plated in a 35 mm μ -Grid dish (Ibidi) a day before microirradiation. Cells were treated with 50 μ M angelicin (ThermoFisher, #501933704) and 1 μ g/ml Hoechst 33342 (Invitrogen, #H3570) for 30 min before transferring to the stage incubator of LSM880 Airyscan confocal laser scanning microscope (Zeiss). Using the bleaching mode in ZEN software, DNA damage was induced by irradiation of a 5-pixel wide region with a 405 nm diode laser (15% power, 1 iteration, zoom 1, averaging 1, pixel dwell time 0.27 μ s = speed 7). One hour after the irradiation, cells were treated with extraction buffer (0.5% Triton X-100, 20 mM Hepes, 50 mM NaCl, 3 mM $MgCl_2$, 300 mM sucrose) for 2 min, washed twice in PBS, fixed in 4% formaldehyde in PBS for 15 min and stained with anti-phospho-Histone H2A.X (Ser139) and anti-SMC6 antibodies following the immunofluorescence protocol.

Immunofluorescence and microscopy

Immunofluorescence and confocal microscopy were done as described (33). For optimal detection of SMC6, we pre-extracted the chromatin with extraction buffer (0.5% Triton X-100, 20 mM Hepes, 50 mM NaCl, 3 mM $MgCl_2$, 300 mM sucrose) for 2 min on ice. Antibodies used for immunofluorescence:

Antibody	Source	Identifier	Dilution
Anti-phospho-Histone H2A.X (Ser139) (Mouse monoclonal)	Sigma	05-636	1:500
Anti-PML (Mouse monoclonal)	Santa Cruz	sc-966	1:200
Anti-SMC6 (Rabbit Polyclonal)	Bethyl	A300-237A	1:500
Anti-SV40 LT (Mouse monoclonal)	Abcam	ab16879	1:400
Goat Anti-mouse IgG (H+L), Alexa Fluor 488 (Goat Polyclonal)	Life Technologies	A11029	1:1000
Goat Anti-rabbit IgG (H+L), Alexa Fluor 488 (Goat Polyclonal)	Life Technologies	A11008	1:1000
Goat Anti-mouse IgG (H+L), Alexa Fluor 555 (Goat Polyclonal)	Life Technologies	A21422	1:1000
Goat Anti-rabbit IgG (H+L), Alexa Fluor 555 (Goat Polyclonal)	Life Technologies	A21428	1:1000

Luciferase and MTT assays

U2OS cells were seeded at 4×10^3 cells per well of a 96-well plate and transfected with 20 ng of luciferase reporter plasmid (expresses secreted Gaussian luciferase under EF1alpha promoter) using TransIT-LT1 (Mirus) by reverse transfection. TAK-981 in indicated concentrations or DMSO were added at the time of transfection and Luciferase activities were measured 3 days later. Ten microliters of culture media were mixed with 50 μ l Gaussian Luciferase Glow Assay buffer that contains freshly diluted coelenterazine substrate (PierceTM Gaussian Luciferase Glow Assay Kit, Thermo Fisher, #16160). Luminescence was measured after ten minutes on a Tecan Infinite M200 plate reader (Tecan Life Sciences). The number of viable cells were estimated using MTT assay kit (Sigma, #M2128) following manufacturer's instructions.

AlphaFold-Multimer Structural modeling

AlphaFold-Multimer modeling ([36](#)) was performed using default settings on the following protein sequences: SIMC1^{Nse5}(425–872), SLF2^{Nse6}(733–1173), and SMC6 (1–1091).

Material Availability

All materials produced during this work are available upon written request, in keeping with the requirements of the journal, funding agencies, and The Scripps Research Institute.

Supplementary figures

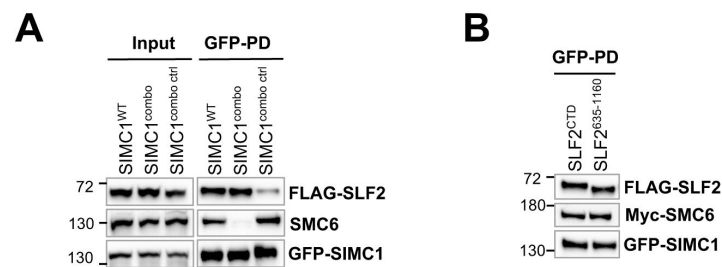


Figure 1 - figure supplement 1.

(A) Western blot of GFP-trap immunoprecipitation from HEK293 cells transiently transfected respectively with GFP-SIMC1, GFP-SIMC1 combo mutant or GFP-SIMC1 combo control mutant in combination with FLAG-SLF2^{CTD} and Myc-SMC6. Signals were detected using FLAG, SMC6 and GFP antibodies. (B) Western blot of GFP-trap immunoprecipitation from HEK293 cells transiently transfected with either FLAG-SLF2^{CTD} or FLAG-SLF2⁶³⁵⁻¹¹⁶⁰ co-transfected with GFP-SIMC1 and Myc-SMC6. Signals were visualized using FLAG, Myc and GFP antibodies.

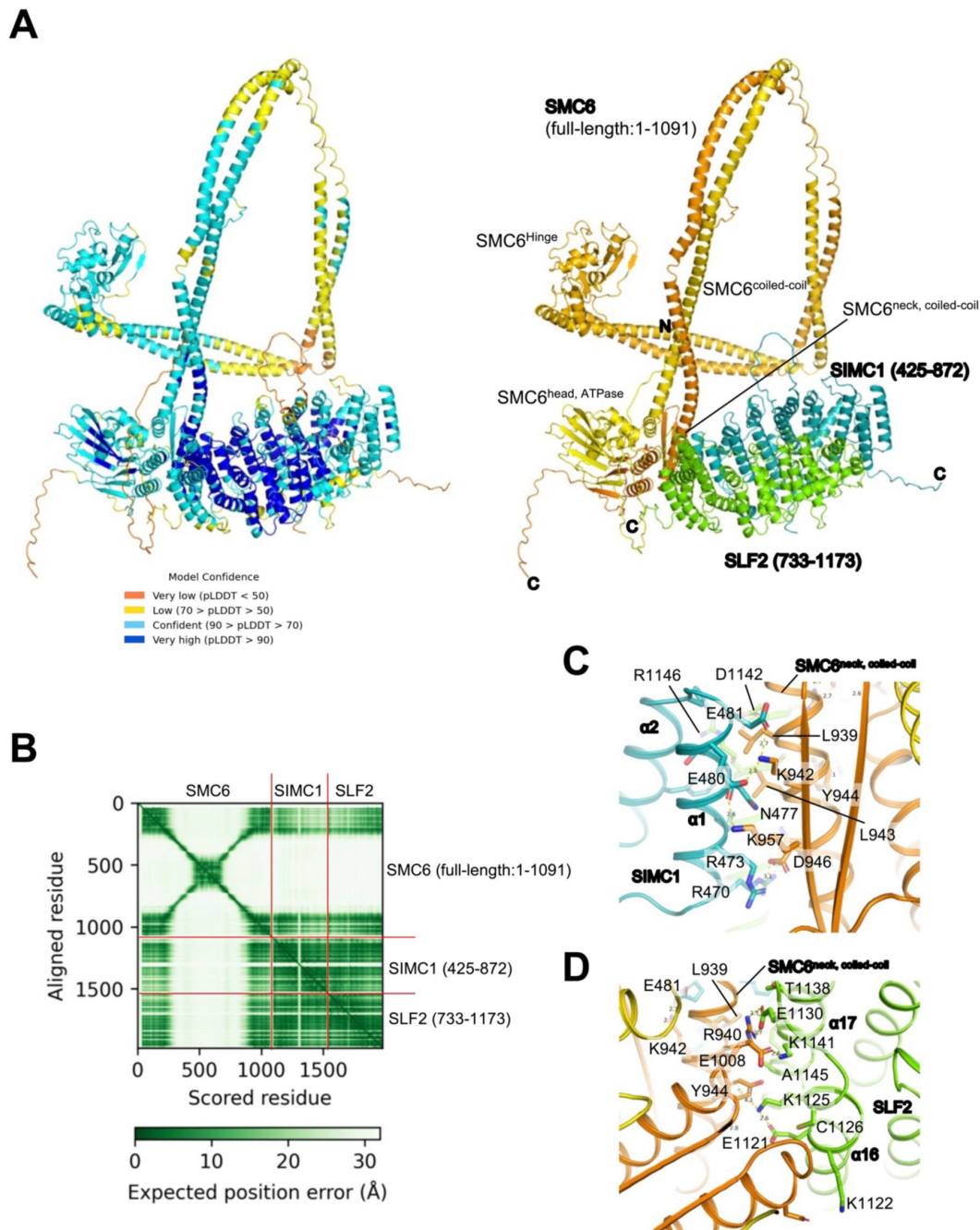


Figure 2 - figure supplement 1

(A) The AlphaFold-Multimer model of the SIMC1^{Nse5} (425-872)-SLF2^{Nse6} (733-1173)-SMC6 (1-1091) complex. The entire model is shown, colored according to pLDDT values (left) or in the same manner as in [Figure 2A](#). (B) The Predicted Aligned Error (PAE) plot of the AlphaFold-Multimer prediction. (C, D) Close-up views of the SMC6 neck-SIMC1 (C) and SMC6 neck-SLF2 (D) interfaces.

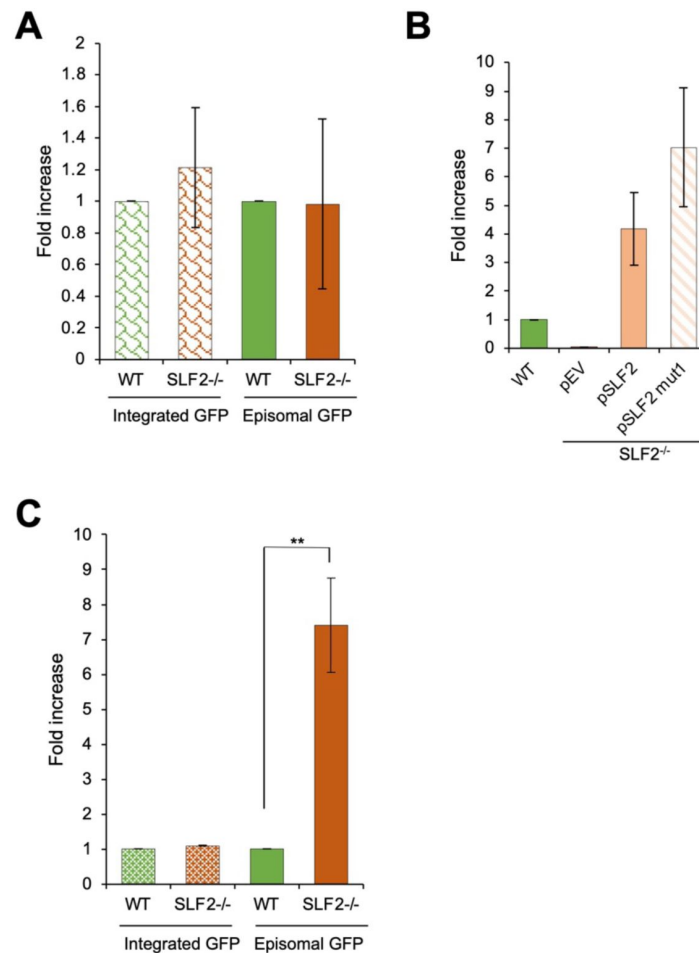


Figure 3 - figure supplement 1

(A) DNA from U2OS WT or SLF2^{-/-} cells harboring either integrated or transiently transfected GFP plasmid was isolated 72 hours after transfection, GFP content was analysed by quantitative PCR and normalized to beta-actin (mean $\pm$ s.d. from n=3 independent experiments, two-tailed unpaired t-test; $p>0.05$). (B) Expression of endogenous and integrated SLF2 WT or SLF2^{mut1} in U2OS WT or SLF2^{-/-} cells was measured by reverse transcription-quantitative PCR and normalized to beta-actin (mean $\pm$ s.d. from n=5 independent experiments). (C) GFP intensity from integrated or transfected GFP vector was measured by flow cytometry 72 hr after reporter transient transfection in U2OS WT cells or SLF2^{-/-} cells. Data are the average $\pm$ s.d. from n=3 independent experiments, two-tailed unpaired t-test; (**) p -value <0.005).

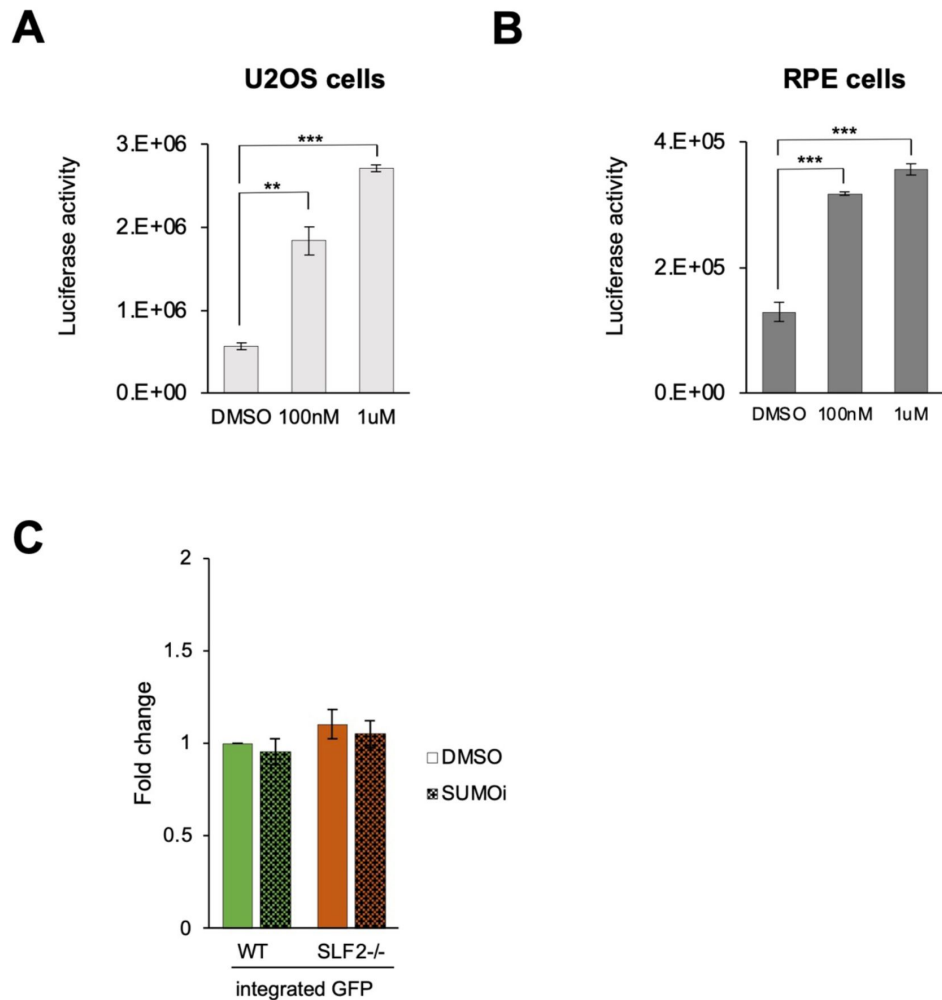


Figure 5 - figure supplement 1

Arbitrary luminescence units of U2OS (**A**) or RPE (**B**) cells expressing transfected pLuc reporter. Cells were treated with TAK-981 or DMSO at the time of transfection. Means and error bars (s.d.) were derived from a minimum of n=3 independent transfections representing biological replicates. Two-tailed unpaired t-test was performed ((**) p<0.005; (***) p<0.0005). (**C**) U2OS wt and SLF2^{-/-} cells stably expressing GFP were treated with 100 nM TAK-981(SUMOi) or DMSO and GFP intensity was determined by FACS 72 hours later. Data normalized to WT cells in DMSO represent the average ± s.d. from n=3 independent experiments.

Figure 5 - figure supplement 2

Representative immunofluorescence images of U2OS WT or SLF2^{-/-} cells fixed and stained with SMC6 (green) and PML (red) antibodies along with DAPI (blue). Scale bar 10 μ m.

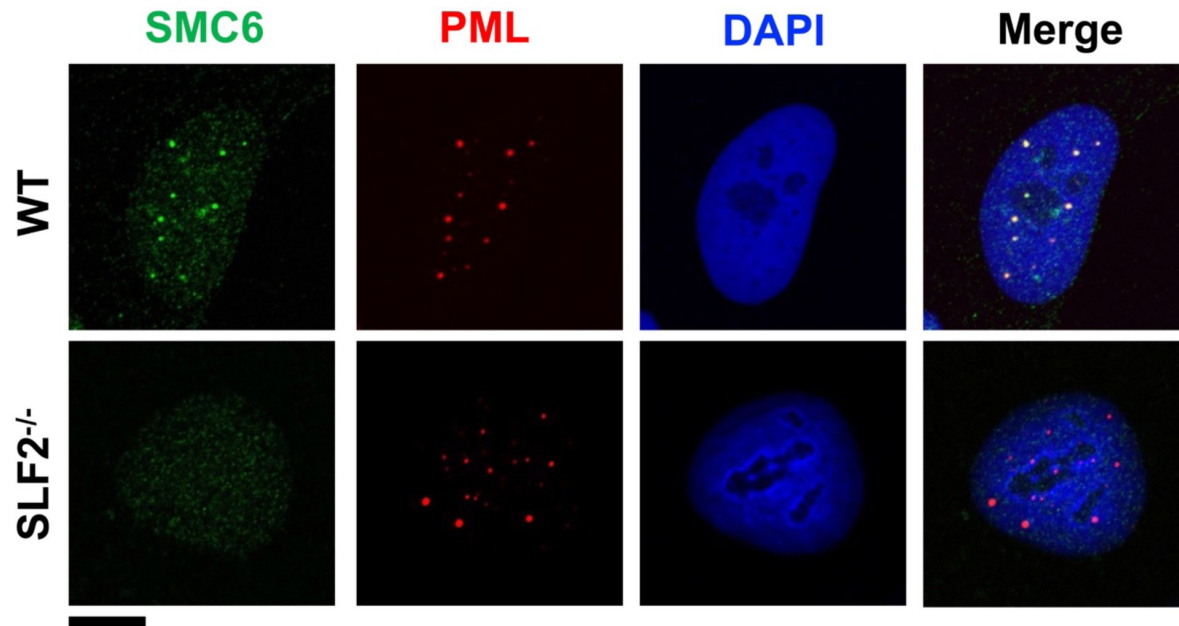
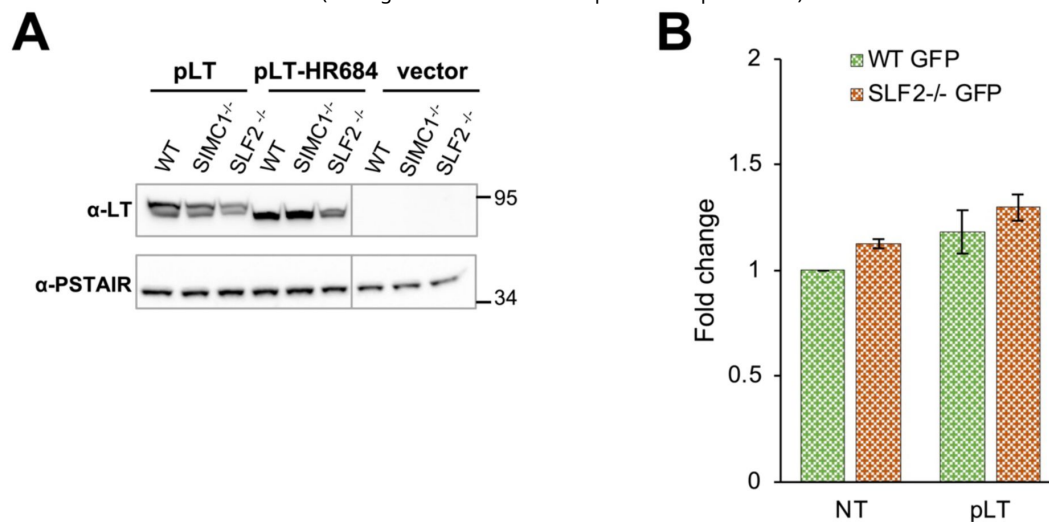


Figure 5 - figure supplement 3

(A) Immunoblot from U2OS WT, SIMC1^{-/-} or SLF2^{-/-} cells stably expressing large T antigen (LT), LT-HR684 variant or have an empty vector integrated, respectively. PSTAIR serves as a loading control. (B) U2OS wt and SLF2^{-/-} cells stably expressing GFP were transiently transfected with vector expressing LT. After 72 hours, GFP intensity was measured by FACS and data normalized to non-transfected WT cells (average $\pm$ s.d. from n=3 independent experiments).



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Author information

Contributions

Conceptualization, M.N.B., T.O. Design of Methodology, M.O., M.N., M.N.B., T.O. Writing – Original Draft, M.N.B., M.N., T.O. Writing – Review & Editing, M.O., M.N., T.O., M.N.B. Formal analysis/Investigation, M.O., M.N., M.N.B., T.O. Validation, M.O., M.N. Resources, M.O., M.N. Visualization, M.O., M.N., T.O., M.N.B. Funding acquisition, M.N.B., T.O. Supervision, M.N.B., T.O. Project administration, M.N.B.

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- Takanori Otomo

Additional files

Figure 1 – source data 1. Full and unedited blots corresponding to panel C. [↗](#)

Figure 2 – source data 1. Full and unedited blots corresponding to panel C. [↗](#)

Figure 3 – source data 1. GFP transcripts qPCR corresponding to panel A. [↗](#)

Figure 3 – source data 2. GFP FACS corresponding to panel B. [↗](#)

Figure 4 – source data 1. GFP FACS corresponding to panel B. [↗](#)

Figure 5 – source data 1. GFP FACS corresponding to panel A. [↗](#)

Figure 5 – source data 2. GFP FACS corresponding to panel B. [↗](#)

Figure 5 – source data 3. Full and unedited blots corresponding to panel C. [↗](#)

Figure 5 – source data 4. GFP FACS corresponding to panel D. [↗](#)

Figure 1 – figure supplement 1 – source data 1. Full and unedited blots corresponding to panel A. [↗](#)

Figure 1 – figure supplement 1 – source data 2. Full and unedited blots corresponding to panel B. [↗](#)

Figure 3 – figure supplement 1 – source data 1. GFP DNA qPCR corresponding to panel A. [↗](#)

Figure 3 – figure supplement 1 – source data 2. SLF2 transcripts qPCR corresponding to panel B. [↗](#)

Figure 3 – figure supplement 1 – source data 3. GFP FACS corresponding to panel C. [↗](#)

Figure 5 – figure supplement 1 – source data 1. Luciferase measurement corresponding to panel A and B. [↗](#)

Figure 5 – figure supplement 1 – source data 2. GFP FACS corresponding to panel C. [↗](#)

Figure 5 – figure supplement 3 – source data 1. Full and unedited blots. [↗](#)

Figure 5 – figure supplement 3 – source data 2. GFP FACS corresponding to panel B. [↗](#)

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Reviewer #1 (Public review):

SMC5/6 is a highly conserved complex able to dynamically alter chromatin structure, playing in this way critical roles in genome stability and integrity that include homologous recombination and telomere maintenance. In the last years, a number of studies have revealed the importance of SMC5/6 in restricting viral expression, which is in part related to its ability to repress transcription from circular DNA. In this context, Oravcova and colleagues recently reported how SMC5/6 is recruited by two mutually exclusive complexes (orthologs of yeast Nse5/6) to SV40 LT-induced PML nuclear bodies (SIMC/SLF2) and DNA lesions (SLF1/2). In this current work, the authors extend this study, providing some new results.

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Reviewer #2 (Public review):

Oravcová et al. present data supporting a role for SIMC1/SLF2 in silencing plasmid DNA via the SMC5/6 complex. Their findings are of interest, and they provide further mechanistic detail of how the SMC5/6 complex is recruited to disparate DNA elements. In essence, the present report builds on the author's previous paper in eLife in 2022 (PMID: 36373674, "The Nse5/6-like SIMC1-SLF2 complex localizes SMC5/6 to viral replication centers") by showing the role of SIMC1/SLF2 in localisation of the SMC5/6 complex to plasmid DNA, and the distinct requirements as compared to recruitment to DNA damage foci.

Reviewer #3 (Public review):

This study by the Boddy and Otomo laboratories further characterizes the roles of SMC5/6 loader proteins and related factors in SMC5/6-mediated repression of extrachromosomal circular DNA. The work shows that mutations engineered at an AlphaFold-predicted protein-protein interface formed between the loader SLF2/SIMC1 and SMC6 (similar to the interface in the yeast counterparts observed by cryo-EM) prevent co-IP of the respective proteins. The mutations in SLF2 also hinder plasmid DNA silencing when expressed in SLF2^{-/-} cell lines, suggesting that this interface is needed for silencing. SIMC1 is dispensable for recruitment of SMC5/6 to sites of DNA damage, while SLF1 is required, thus separating the functions of the two loader complexes. Preventing SUMOylation (with a chemical inhibitor) increases transcription from plasmids but does not in SLF2-deleted cell lines, indicating the SMC5/6 silences plasmids in a SUMOylation dependent manner. Expression of LT is sufficient for increased expression, and again, not additive or synergistic with SIMC1 or SLF2 deletion, indicating that LT prevents silencing by directly inhibiting 5/6. In contrast, PML bodies appear dispensable for plasmid silencing.

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Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):

SMC5/6 is a highly conserved complex able to dynamically alter chromatin structure, playing in this way critical roles in genome stability and integrity that include homologous recombination and telomere maintenance. In the last years, a number of studies have revealed the importance of SMC5/6 in restricting viral expression, which is in part related to its ability to repress transcription from circular DNA. In this context, Oravcova and colleagues recently reported how SMC5/6 is recruited by two mutually exclusive complexes (orthologs of yeast Nse5/6) to SV40 LT-induced PML nuclear bodies (SIMC/SLF2) and DNA lesions (SLF1/2). In this current work, the authors extend this study, providing some new results. However, as a whole, the story lacks unity and does not delve into the molecular mechanisms responsible for the silencing process. One has the feeling that the story is somewhat incomplete, putting together not directly connected results.

Please see the introductory overview above.

(1) In the first part of the work, the authors confirm previous conclusions about the relevance of a conserved domain defined by the interaction of SIMC and SLF2 for their binding to SMC6, and extend the structural analysis to the modelling of the SIMC/SLF2/SMC complex by AlphaFold. Their data support a model where this conserved surface of SIMC/SLF2 interacts with SMC at the backside of SMC6's head domain, confirming the relevance of this interaction site with specific mutations. These results are interesting but confirmatory of a previous and more complete structural analysis in yeast (Li et al. NSMB 2024). In any case, they reveal the conservation of the interaction. My major concern is the lack of connection with the rest of the article. This structure does not help to understand the process of transcriptional silencing reported later beyond its

relevance to recruit SMC5/6 to its targets, which was already demonstrated in the previous study.

Demonstrating the existence of a conserved interface between the Nse5/6-like complexes and SMC6 in both yeast and human is foundationally important, not confirmatory, and was not revealed in our previous study. It remains unclear how this interface regulates SMC5/6 function, but yeast studies suggest a potential role in inhibiting the SMC5/6 ATPase cycle. Nevertheless, the precise function of Nse5/6 and its human orthologs in SMC5/6 regulation remain undefined, largely due to technical limitations in available *in vivo* analyses. The SIMC1/SLF2/SMC6 complex structure likely extends to the SLF1/2/SMC6 complex, suggesting a unifying function of the Nse5/6-like complexes in SMC5/6 regulation, albeit in the distinct processes of ecDNA silencing and DNA repair. There have been no studies to date (including this one) showing that SIMC1-SLF2 is required for SMC5/6 recruitment to ecDNA. Our previous study showed that SIMC1 was needed for SMC5/6 to colocalize with SV40 LT antigen at PML NBs. Here we show that SIMC1 is required for ecDNA repression, in the absence of PML NBs, which was not anticipated.

(2) In the second part of the work, the authors focus on the functionality of the different complexes. The authors demonstrate that SMC5/6's role in transcription silencing is specific to its interaction with SIMC/SLF2, whereas SMC5/6's role in DNA repair depends on SLF1/2. These results are quite expected according to previous results. The authors already demonstrated that SLF1/2, but not SIMC/SLF2, are recruited to DNA lesions. Accordingly, they observe here that SMC5/6 recruitment to DNA lesions requires SLF1/2 but not SIMC/SLF2. Likewise, the authors already demonstrated that SIMC/SLF2, but not SLF1/2, targets SMC5/6 to PML NBs. Taking into account the evidence that connects SMC5/6's viral resistance at PML NBs with transcription repression, the observed requirement of SIMC/SLF2 but not SLF1/2 in plasmid silencing is somehow expected. This does not mean the expectation has not to be experimentally confirmed. However, the study falls short in advancing the mechanistic process, despite some interesting results as the dispensability of the PML NBs or the antagonistic role of the SV40 large T antigen. It had been interesting to explore how LT overcomes SMC5/6-mediated repression: Does LT prevent SIMC/SLF2 from interacting with SMC5/6? Or does it prevent SMC5/6 from binding the plasmid? Is the transcription-dependent plasmid topology altered in cells lacking SIMC/SLF2? And in cells expressing LT? In its current form, the study is confirmatory and preliminary. In agreement with this, the cartoons modelling results here and in the previous work look basically the same.

Our previous study only examined the localization of SLF1 and SIMC1 at DNA lesions. The localization of these subcomplexes alone should not be used to define their roles in SMC5/6 localization. Indeed, the field is split in terms of whether Nse5/6-like complexes are required for ecDNA binding/loading, or regulation of SMC5/6 once bound.

We agree, determining the potential mechanism of action of LT in overcoming SMC5/6-based repression is an important next step. We believe it is unlikely due to blocking of the SMC5/6SIMC1/SLF2 interface, since SIMC1-SLF2 is required for SMC5/6 to localize at LT-induced foci. It will require the identification of any direct interactions with SMC5/6 subunits, and better methods for assessing SMC5/6 loading and activity on ecDNAs. Unlike HBx, Vpr, and BNRF1 it does not appear to induce degradation of SMC5/6, making it a more complex and interesting challenge. Also, the dispensability of PML NBs in plasmid silencing versus viral silencing raises multiple important questions about SMC5/6's repression mechanism.

(3) There are some points about the presented data that need to be clarified.

Thank you, we have addressed these points below, within the Recommendations for authors section.

Reviewer #2 (Public review):

Oracová et al. present data supporting a role for SIMC1/SLF2 in silencing plasmid DNA via the SMC5/6 complex. Their findings are of interest, and they provide further mechanistic detail of how the SMC5/6 complex is recruited to disparate DNA elements. In essence, the present report builds on the author's previous paper in eLife in 2022 (PMID: 36373674, "The Nse5/6-like SIMC1-SLF2 complex localizes SMC5/6 to viral replication centers") by showing the role of SIMC1/SLF2 in localisation of the SMC5/6 complex to plasmid DNA, and the distinct requirements as compared to recruitment to DNA damage foci. Although the findings of the manuscript are of interest, we are not yet convinced that the new data presented here represents a compelling new body of work and would better fit the format of a "research advance" article. In their previous paper, Oracová et al. show that the recruitment of SMC5/6 to SV40 replication centres is dependent on SIMC1, and specifically, that it is dependent on SIMC1 residues adjacent to neighbouring SLF2.

We agree. We submitted this manuscript as a "Research Advance", not as a standalone research article, given that it is an extension of our previous "Research Article" (1).

Other comments

(1) The mutations chosen in Figure 1 are quite extensive - 5 amino acids per mutant. In addition, they are in many cases 'opposite' changes, e.g., positive charge to negative charge. Is the effect lost if single mutations to an alanine are made?

The mutations were chosen to test and validate the predicted SIMC1-SLF2-SMC6 structure i.e. the contact point between the conserved patch of SIMC1-SLF2 and SMC6. Multiple mutations and charge inversions increased the chance of disrupting the extensive interface. In this respect, the mutations were successful and informative, confirming the requirement of this region in specifically contacting SMC6. Whilst alanine scanning mutations are possible, we believe that they would not add to, or detract from, our validation of the predicted SIMC1-SLF2-SMC6 interface.

(2) In Figure 2c, it isn't clear from the data shown that the 'SLF2-only' mutations in SMC6 result in a substantial reduction in SIMC1/SLF2 binding.

To clarify the difference between wild-type and SLF2-only mutations in SIMC1-SLF2 interaction, we have performed an image volume analysis. This shows that the SLF2-facing SMC6 mutant reduces its interaction with SIMC1 (to 44% of WT) and SLF2 (to 21% of WT). The reduction in both SIMC1 and SLF2 interaction with SMC6 SLF2-facing mutant is expected, since SIMC1 and SLF2 are an interdependent heterodimer.

Author response table 1.

mutant	%
GFP-SMC6 blot	
wt-SMC6	100
SLF2 only	120,0891
Myc-SMC1 blot	
wt-SMC6	100
SLF2 only	43,88612
Flag-SLF2 blot	
wt-SMC6	100
SLF2 only	21,28111

(3) In the GFP reporter assays (e.g. Figure 3), median fluorescence is reported - was there any observed difference in the percentage of cells that are GFP positive?

Yes, as expected when the GFP plasmid is not actively repressed, the percent of GFP positive cells differs in each cell line – in the same trend as GFP intensity

(4) The potential role of the large T antigen as an SMC5/6 evasion factor is intriguing. However, given the role of the large T antigen as a transcriptional activator, caution is required when interpreting enhanced GFP fluorescence. Antagonism of the SMC5/6 complex in this context might be further supported by ChIP experiments in the presence or absence of large T. Can large T functionally substitute for HBx or HIV-Vpr?

We agree, the potential role of LT in SMC5/6 antagonism is interesting. We did state in the text “While LT is known to be a promiscuous transcriptional activator (2,3) that does not rule out a co-existing role in antagonizing SMC5/6. Indeed, these findings are reminiscent of HBx from HBV and Vpr of HIV-1, both of which are known promiscuous transcriptional activators that also directly antagonize SMC5/6 to relieve transcriptional repression (4-10).” We have tried ChIP experiments, but found these to be unreliable in assessing SMC5/6 association with plasmid DNA. Given the many disparate targets of LT, HBx and Vpr (other than SMC5/6), it seems unlikely that LT could functionally substitute for HBx and Vpr in supporting HBV and HIV-1 infections. Whilst certainly an interesting future question, we believe it is beyond the scope of this study.

(5) In Figure 5c, the apparent molecular weight of large T and SMC6 appears to change following transfection of GFP-SMC5 - is there a reason for this?

We are not certain as to what causes the molecular weight shift, but it is not specifically related to GFP-SMC5 transfection. Rather, it appears to be a general effect of the pulldown. Indeed, a very weak “background” band of LT is seen in the GFP only pulldown, which also runs at a “higher” molecular weight, as in the GFP-SMC5 pulldown. We believe that the effect is instead related to gel mobility in the wells that contain post pulldown proteins and different buffers. We have also seen similar effects using different protein-protein interaction pairs.

Reviewer #3 (Public review):

Summary:

This study by the Boddy and Otomo laboratories further characterizes the roles of SMC5/6 loader proteins and related factors in SMC5/6-mediated repression of extrachromosomal circular DNA. The work shows that mutations engineered at an AlphaFold-predicted protein-protein interface formed between the loader SLF2/SIMC1 and SMC6 (similar to the interface in the yeast counterparts observed by cryo-EM) prevent co-IP of the respective proteins. The mutations in SLF2 also hinder plasmid DNA silencing when expressed in SLF2-/- cell lines, suggesting that this interface is needed for silencing. SIMC1 is dispensable for recruitment of SMC5/6 to sites of DNA damage, while SLF1 is required, thus separating the functions of the two loader complexes. Preventing SUMOylation (with a chemical inhibitor) increases transcription from plasmids but does not in SLF2-deleted cell lines, indicating the SMC5/6 silences plasmids in a SUMOylation dependent manner. Expression of LT is sufficient for increased expression, and again, not additive or synergistic with SIMC1 or SLF2 deletion, indicating that LT prevents silencing by directly inhibiting 5/6. In contrast, PML bodies appear dispensable for plasmid silencing.

Strengths:

The manuscript defines the requirements for plasmid silencing by SMC5/6 (an interaction of Smc6 with the loader complex SLF2/SIMC1, SUMOylation activity) and shows that SLF1 and PML bodies are dispensable for silencing. Furthermore, the authors show that LT can overcome silencing, likely by directly binding to (but not degrading) SMC5/6.

Weaknesses:

(1) Many of the findings were expected based on recent publications.

There have been no manuscripts describing the role of SIMC1-SLF2 in ecDNA silencing. There have been studies describing SLF2's roles in ecDNA silencing, but these suggested SLF2 had an SLF1 independent role, with no mention of an alternate Nse5-like cofactor. Our earlier study in eLife (1) described the identification of SIMC1 as an Nse5-like cofactor for SLF2 but did not test potential roles of the complex in ecDNA silencing. Also, the apparent dispensability of PML NBs in plasmid silencing (in U2OS cells) was unexpected based on recent publications. Finally, SV40 LT has not previously been implicated in SMC5/6 inhibition, which may occur through novel mechanisms.

(2) While the data are consistent with SIMC1 playing the main function in plasmid silencing, it is possible that SLF1 contributes to silencing, especially in the absence of SIMC1. This would potentially explain the discrepancy with the data reported in ref. 50. SLF2 deletion has a stronger effect on expression than SIMC1 deletion in many but not all experiments reported in this manuscript. A double mutant/deletion experiments would be useful to explore this possibility.

It is interesting to note that the data in ref. 50 (11) is also at odds with that in ref. 45 (8) in terms of defining a role for SLF1 in the silencing of unintegrated HIV-1 DNA. The Irwan study showed that SLF1 deficient cells exhibit increased expression of a reporter gene from unintegrated HIV-1, whereas the Dupont study found that SLF1 deletion, unlike SLF2 deletion, has no effect. It is unclear what the basis of this discrepancy is. In line with the Dupont study, we found no effect of SLF1 deletion on plasmid expression (Figure 4B), whereas SLF2 deletion increased reporter expression (Figure 3A/B). It is possible that SLF1 could support some plasmid silencing in the absence of SIMC1, especially considering the gross structural similarity in their C-terminal Nse5-like domains. However, we have been unable to generate double-knockout SIMC1 and SLF1 cells to test such a possibility, and shSLF1 has been ineffective.

(3) SLF2 is part of both types of loaders, while SLF1 and SIMC1 are specific to their respective loaders. Did the authors observe differences in phenotypes (growth, sensitivities to DNA damage) when comparing the mutant cell lines or their construction? This should be stated in the manuscript.

We have not observed significant differences in the growth rates of each cell line, and DNA damage sensitivities are as yet untested.

(4) It would be desirable to have control reporter constructs located on the chromosome for several experiments, including the SUMOylation inhibition (Figures 5A and 5-S2) and LT expression (Figure 5D) to exclude more general effects on gene expression.

We have repeated all GFP reporter assays using integrated versus episomal plasmid DNA. A seminal study by Decorsière et al. (6) showed that SMC5/6 degradation by HBx of HBV increased transcription of episomal but not chromosomally integrated reporters. In line with this data, the deletion of SLF2 does not notably impact the expression of our GFP reporter construct when it is genomically integrated (Figure 3—figure supplement 1C).

Somewhat surprisingly, given the generally transcriptionally repressive roles of SUMO, inhibition of the SUMO pathway with SUMOi did not significantly impact the expression of our genomically integrated GFP reporter, versus the episomal plasmid (Figure 5—figure supplement 1C). Finally, the expression of SV40 LT, which enhances plasmid reporter expression (Figure 5D), also did not notably affect expression of the same reporter when located in the genome (Figure 5—figure supplement 3B). This is an interesting result, which is in line with an early study showing that HBx of HBV induces transcription from episomal, but not chromosomally integrated reporters (12). This further suggests that SV40 LT acts similarly to other early viral proteins like HBx and Vpr to counteract or bypass SMC5/6 restriction, amongst their multifaceted functions. Clearly, further analyses are needed to define mechanisms of LT in counteracting SMC5/6, but they do not appear to include complex degradation as seen with HBx and Vpr.

(5) Figure 5A: There appears to be an increase in GFP in the SLF2^{-/-} cells with SUMOi? Is this a significant increase?

No significant difference was found between WT, SIMC1^{-/-} or SLF2^{-/-} when treated with SUMOi ($p > 0.05$). The p-value is 0.0857 (when comparing SLF2^{-/-} to WT in the SUMOi condition) This is described in the figure legend to Figure 5.

(6) The expression level of SFL2 mut1 should be tested (Figure 3B).

Full length SLF2 (WT or mutants) has been undetectable by western analyses. However, truncated SLF2 mut1 expresses well and binds SIMC1 but not SMC6 (Figure 1C). Moreover, full length SLF2 mut1 expression was confirmed by qPCR – showing a somewhat higher expression level than SLF2 WT (Figure 3—figure supplement 1B).

Reviewer #1 (Recommendations for the authors):

There are some points about the presented data that need to be clarified.

(1) Figures 3, 4B, and 5. The authors should rule out the possibility that the reported effects on transcription were due to alterations in plasmid number. This is particularly important, taking into account the importance of SMC5/6 in DNA replication.

We used qPCR to assess plasmid copy number versus genomic DNA in our cell lines, testing at 72 hours post transfection to avoid any impact of cytosolic DNA (13). Our qPCR data show that there is no significant impact on plasmid copy number across our cell lines i.e. WT and SLF2 null. SMC5/6 has a positive role in DNA replication progression on the genome (e.g. (14)), so loss of SMC5/6 “targeting” in SIMC1 and SLF2 null cells would be unlikely to promote replication fork progression per se.

(2) Figure S1A. In contrast to the statement in the text, the SIMC1-combo control is affected in its binding to SLF2; however, it is not affected in its binding to SMC6. This is somehow unexpected because it suggests that the solenoid-like structure is not required for SMC6 binding, just specific patches at either SIMC or SLF2. This should be commented on.

We appreciate the reviewer’s observation regarding the discrepancy between Figure S1A and the text. This was our oversight. The data show that SLF2 recovery was reduced in the pull-down with the SIMC1 combo control mutant, while SLF2 expression was unchanged. Because SLF2 or SIMC1 variants that fail to associate typically show poor expression (1), these findings suggest that the SIMC1 combo control mutant associates with SLF2, albeit more weakly. Since the mutations were introduced into surface residues of SIMC1, it is not immediately clear how they would weaken the interaction or destabilize the complex. In contrast, SMC6 was fully recovered with the SIMC1 combo control mutant, indicating that the SIMC1–SMC6 interaction remains stable without stoichiometric SLF2. This may reflect direct recognition of a SIMC1 binding epitope or stabilization of its solenoid structure by SMC6, although this interpretation remains uncertain given the unstable nature of free SIMC1 and SLF2. Alternatively, SMC6 may have co-sedimented with the SIMC1 combo control mutant together with SLF2, which was initially retained but subsequently lost during washing, whereas SMC6 remained due to its limited solubility in the absence of other SMC5/6 subunits. While further mechanistic analysis will require purified SMC5/6 components, our data support the AlphaFold-based model by demonstrating that SIMC1 mutations on the non–SMC6-contacting surface retain association with SMC6. The text has been revised accordingly.

(3) The SLF2-only mutant has alterations that affect interactions with both SLF2 and SIMC1. Is it not another Mixed mutant?

We appreciate the reviewer’s observation regarding the discrepancy between the mutant name (“SLF2only”) and its description (“while N947 forms salt bridges with SIMC1”). The previous statement was inaccurate due to a misinterpretation of several AlphaFold models. Across these models, the SIMC1–SLF2 interface residues remain largely consistent, but the SIMC1 residue R470 exhibits positional variability—contacting N947 in some models but not in others. Given this variability and the absence of an experimental structure, we have revised the text to avoid overinterpretation. Because the N947 side chain is oriented toward SLF2 and consistently forms polar contacts with the H1148 side chain and G1149 backbone, we have renamed this mutant “SLF2-facing,” which more accurately describes its modeled environment. The other mutants are likewise renamed “SIMC1-facing” and “SIMC1–SLF2groove-facing,” providing a clearer and more consistent description of the interface.

(4) The SLF2-only mutant still displays clear interactions with SMC6. Can this be explained with the AlphaFold model?

SIMC1 may contribute more substantially to SMC6 binding than SLF2, consistent with our mutagenesis results. However, the energetic contributions of individual residues or proteins cannot be quantitatively inferred from structural models alone. Comprehensive experimental and computational analyses would be required to address this point.

(5) *The conclusions about the role of SUMOylation are vague; it is already known that its general effect on transcription repression, and the authors already demonstrated that SIMC interacts with SUMO pathway factors. Concerning the epistatic effect, the experiment should be done at a lower inhibitor concentration; at 100 nM there is not much margin to augment according to the kinetics analysis in Figure S5.*

The SUMO pathway is indeed thought to be generally repressive for transcription. Notably, in response to a suggestion from Reviewer 3 (public review point 4), we have repeated several of our GFP expression assays using cells with the GFP reporter plasmid integrated into the genome (please see Figure 3—figure supplement 1C; Figure 5—figure supplement 1C; Figure 5—figure supplement 3B). This type of integrated reporter does not show elevated expression following inhibition of the SMC5/6 complex, unlike ecDNAs (6,10). Interestingly, SUMOi, LT expression, and SLF2 knockout also did not notably impact the expression of our integrated GFP reporter (Figure 3—figure supplement 1C; Figure 5—figure supplement 1C; Figure 5—figure supplement 3B, unlike that of the plasmid (ecDNA) reporter. Given the “general” inhibitory effect of SUMO on transcription, the SUMOi result was not expected, and it opens further interesting avenues for study.

In Figure 5—figure supplement 1A, 100 nM SUMOi increases reporter expression well below the highest SUMOi dose. We believe that the ~3-4 fold induction of GFP expression in SLF2 null cells, if independent of SUMOylation, should further increase GFP expression. The impact of SUMOylation on GFP reporter expression remains “vague”, but our data indicate that SMC5/6 operates within SUMO’s “umbrella” function and provides a starting point for more mechanistic dissection.

(6) *Figure 5C. Why is the size different between Input versus GFP-PD?*

Please see our response to this question above: reviewer 2, point (5)

Reviewer #2 (Recommendations for the authors):

If further data could be provided to extend on that which is presented, then publication as a 'standalone research article' may be appropriate, but not in its present form.

We submitted this manuscript as a “Research Advance” not as a standalone research article, given that it was an extension of our previous research article (1).

Reviewer #3 (Recommendations for the authors):

(1) *The term 'LT' should be defined in the title*

We have updated the title accordingly.

(2) *This reviewer found the nomenclature of the SMC6 mutants confusing (SIMC1-only...). Either rephrase or define more clearly in the text and the figures.*

We agree with the reviewer and have renamed the mutants as “SIMC1-facing”, “SLF2-facing”, and “SIMC1-SLF2-groove-facing”.

(3) *The authors could better emphasize that LT blocks silencing in trans (not only on its cognate target sequence in cis). This is consistent with the observed direct binding to SMC5/6.*

We appreciate the suggestion to further emphasize the impact of LT on plasmid silencing. We did not want to overstate its impact at this time because we do not know if it directly binds SMC5/6 or indeed affects SMC5/6 function more broadly. LT expression like HBx, does cause induction of a DNA damage response, but we cannot at this point tie that response to SMC5/6 inhibition alone.

(4) Figure 5 S1: the merge looks drastically different. Is DAPI omitted in the wt merge image?

Thank you for noting this issue. We have corrected the image, which was impacted by the use of an underexposed DAPI image.

(5) Figure 1: how is the structure in B oriented relative to A? A visual guide would be helpful.

We have added arrows to indicate the view orientation and rotational direction to turn A to B.

(6) Line 126, unclear what "specificity" here means.

We have revised the sentence without this word, which now starts with "To confirm the SIMC1-SMC6 interface, we introduced...."

(7) Line 152, The statement implies that the conserved residues are needed for loader subunits interactions ('mediating the SIMC1-SLF2 interaction'). Does Figure 1C not show that the residues are not important? Please clarify.

Thank you for noting this writing error. We have corrected the sentence to provide the intended meaning. It now reads "Collectively, these results confirm that the conserved surface patch of SIMC1SLF2 is essential for SMC6 binding."

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